

Phases of Natural Language Processing

A Comprehensive Guide from Sound to Meaning

Introductory Course on NLP Pipeline

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Learning Objectives



Understand Phases

Identify and define the 6 primary phases of the NLP pipeline from input to output.



Learn Mechanisms

Understand the specific tasks, inputs, outputs, and common challenges within each step.



System Integration

See how all phases work together sequentially to derive real meaning from human language.

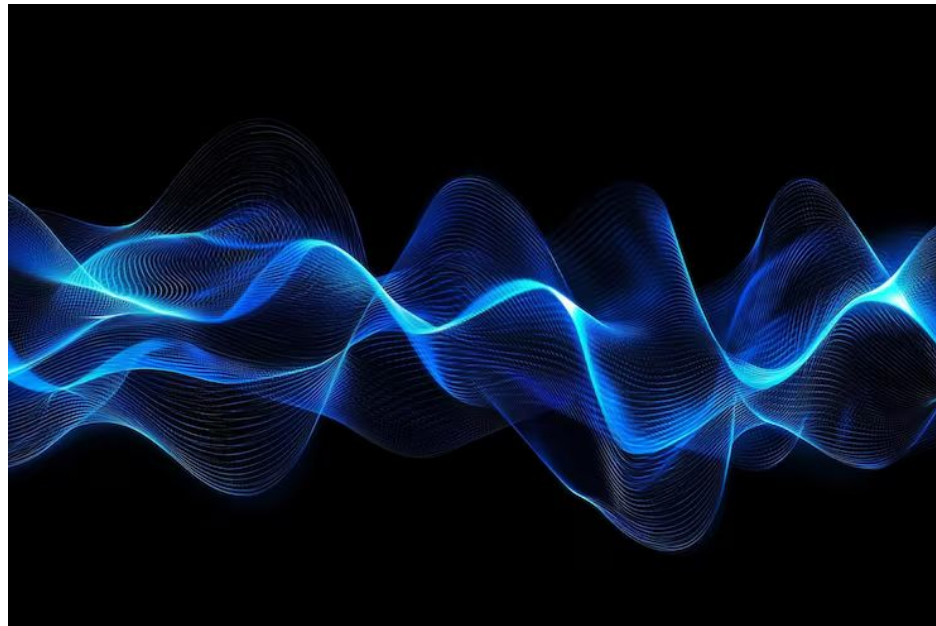
Instructor Note: Welcome students. Explain that NLP is not a single black box, but a step-by-step pipeline. Compare it to human comprehension: we hear sounds, form words, structure sentences, understand meaning, and apply context. We will walk through this exact biological process as it applies to machines.

1. Phonological Analysis

Definition: The study of sounds in language. It converts physical sound waves into distinct acoustic sounds (phonemes).

Purpose: Crucial for spoken language systems to digitize human voice.

- **Input:** Audio/Speech waveform.
- **Output:** Text string / Phonetic representation.
- **Real-world Application:** Siri, Google Assistant, Voice-to-Text.
- **Common Challenge:** Accents, background noise, homophones (e.g., "write" vs. "right").



Instructor Note: Emphasize that this phase is ONLY required for spoken language (Speech Recognition). If the input is already text (like a typed search query), the NLP system completely skips this phase and starts at Lexical Analysis.

2. Lexical and Morphological Analysis

Definition: Identifying and analyzing the structure of words. The text is broken down into smaller chunks.

Key Concepts:

- **Tokenization:** Breaking a sentence into individual words or punctuation marks (tokens).
- **Morphological analysis** studies the internal structure of words using morphemes, the smallest units of meaning.
 - Example: Stemming and Lemmatization
- **Morphemes:** The smallest meaningful units (e.g., "un-break-able").

Input: Raw text sequence.

Output: Stream of tokens/words.

Instructor Note: Provide a quick example on the board: The sentence "I am running" becomes three tokens: ["I", "am", "running"]. Explain that a common mistake here is mishandling punctuation (e.g., "U.S.A." vs "U S A").

3. Syntactic Analysis (Parsing)

What is Syntax?

Definition: Analyzing the structural grammar of the sentence. It ensures the tokens follow the grammatical rules of the language and helps identify the relationships between words.

POS Tagging: assigns grammatical roles such as noun, verb or adjective to words.

Process: The system creates a **Parse Tree**, mapping nouns, verbs, and adjectives to understand structural relationships.

Output: A structurally validated tree representing the sentence.

The Ambiguity Challenge

Example 1: *"I saw the man with a telescope."*

A machine faces **Syntactic Ambiguity** here:

- Meaning A: I used a telescope to see the man.
- Meaning B: I saw a man who was holding a telescope.

Syntax checks grammar, but cannot always resolve structural double-meanings alone.

4. Semantic Analysis

Extracting Literal Meaning

Definition: Focusing on the exact, literal, dictionary meaning of words and how they combine to form a meaningful sentence.

Purpose: To ensure that text is not only grammatically correct but also logically meaningful and contextually appropriate.

Key Tasks:

- **Named Entity Recognition(NER):** Identifies entities such as people, location, organizations and dates.
- **Word Sense Disambiguation (WSD):** Determines the correct meaning of words with multiple meanings based on context

Example:

- **Sentence: Apple eats a John.**
- Although the sentence is grammatically correct, it is semantically incorrect because an apple cannot perform the action of eating a person.

5. Discourse Analysis

Definition: Analyzing how sentences relate to one another in a larger block of text or conversation.

Key Concept (Coreference Resolution): Determining which words ("he", "she", "it") refer to which entities in previous sentences.

Example:

- Person A: "*John* bought a new *car*."
- Person B: "*He* loves *it*."

The system maps "He" back to "John", and "it" back to "car". Without Discourse Analysis, the second sentence is meaningless.

Key Aspects:

- **Anaphora Resolution:** Identifies references such as pronouns and links them to the correct subject.
- **Contextual References:** Understands words or phrases based on surrounding context.

6. Pragmatic Analysis (Context matters)

Semantic Meaning (Phase 4)

Focuses on **literal interpretation**.

Input: "Can you open the window?"

System Interpretation: An inquiry about the listener's physical capacity or permission to perform the action of opening a window.

Expected Output: "Yes, I am physically able to."

Pragmatic Meaning (Phase 5)

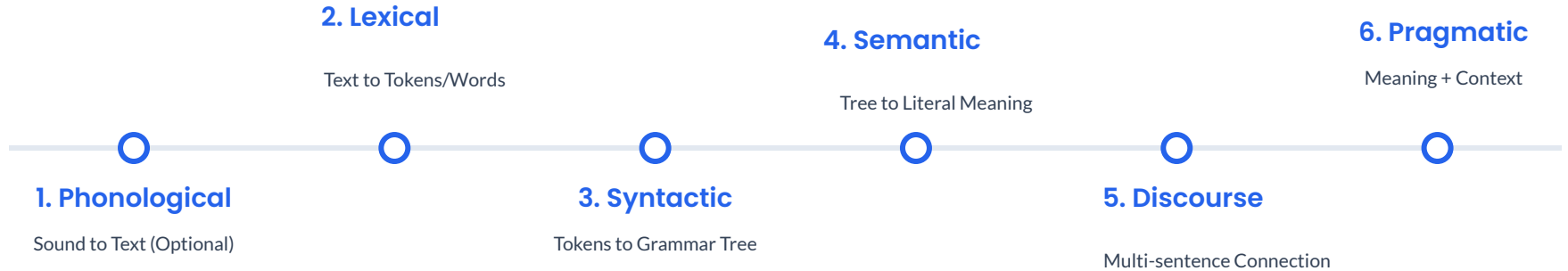
Focuses on **intended context and real-world use**.

Input: "Can you open the window?"

System Interpretation: A polite command or request for the listener to open the window because the speaker is likely hot or needs air.

Expected Output: Action (opening the window).

Phases of NLP



Phase Comparison Table

NLP Phase	Main Goal	Input	Output
Phonological	Interpret spoken sounds	Audio / Voice	Raw text sequence
Lexical	Identify words/morphemes	Raw text	Tokens (word lists)
Syntactic	Verify grammatical structure	Tokens	Parse Tree
Semantic	Extract literal dictionary meaning	Parse Tree	Meaning representation
Pragmatic	Determine contextual intent	Literal Meaning	Intended Action/Meaning
Discourse	Link sentences together	Current Sentence	Cross-sentence context

Key Takeaways

- **Hierarchy of Understanding:** NLP moves from the smallest units (sounds/characters) to the largest structures (entire conversations).
- **Ambiguity is the Enemy:** Every phase exists to eliminate a different type of ambiguity (audio, grammatical, literal, contextual).
- **Syntax != Semantics:** A sentence can be perfectly grammatical but completely meaningless. Both phases are required.

FAQs

Q1: What is Tokenization?

A: The process in the Lexical phase where raw text is split into smaller units (tokens) like words, subwords, or punctuation for the machine to process.

Q2: Differentiate Syntax & Semantics.

A: Syntax ensures the sentence follows grammatical rules (structure). Semantics ensures the sentence actually makes logical sense (meaning).

Q3: What is a Parse Tree?

A: A hierarchical tree structure produced during Syntactic Analysis that represents the grammatical structure of a sentence (Nouns, Verbs, etc).

Q4: Give an example of Pragmatics.

A: "Do you have a watch?" Literal meaning (Semantic) is asking about ownership. Pragmatic meaning is asking "What time is it?".

FAQs

Q5: What is Coreference Resolution?

A: A task in Discourse Analysis where the system identifies all expressions in a text that refer to the same entity (e.g., matching "she" to "Mary").

Q6: Why is NLP difficult?

A: Because human language is highly ambiguous, unstructured, relies on cultural context, and is full of idioms and sarcasm.

Q7: What phase processes Morphemes?

A: Lexical Analysis. It breaks words into root forms and affixes (e.g., "un-happi-ness").

Q8: If I type a query into Google, which phase is skipped?

A: Phonological Analysis. Since the input is already text, there are no audio waves to process.

Knowledge Check (Classroom Quiz)

1. Which phase handles "He" referring back to "John"?

A) Lexical Analysis

B) Discourse Analysis (Correct)

2. Breaking a sentence into distinct words is called:

A) Tokenization (Correct)

B) Syntactic Parsing

3. Resolving the dual-meaning of "I saw the man with a telescope" requires:

A) Phonological Analysis

B) Syntactic/Semantic Analysis (Correct)

4. Understanding that "It's cold in here" actually means "close the door" is:

A) Semantics

B) Pragmatics (Correct)

Thank you!